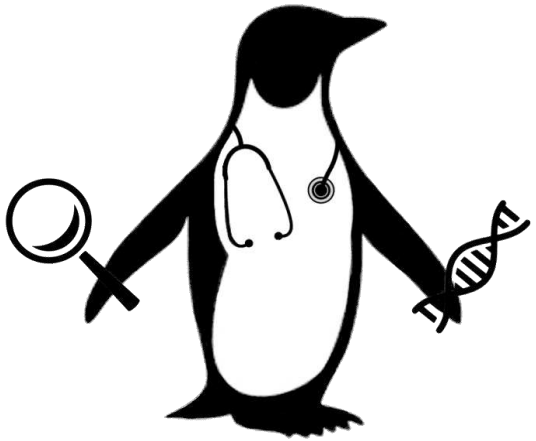


PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice

Elias Crum

European Semantic Web Conference
Ph.D. Symposium (Early Stage)
May 27th, 2024

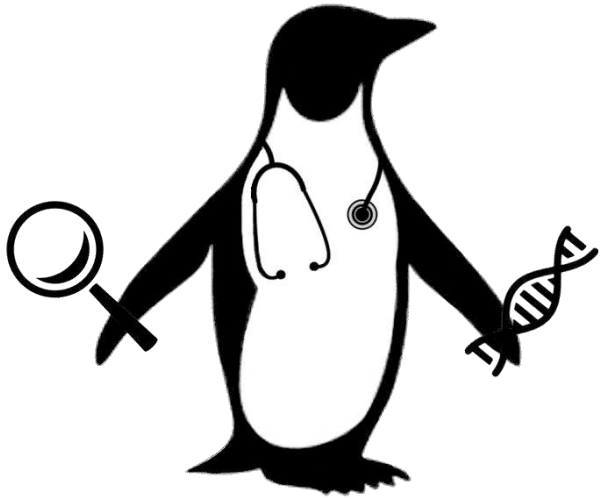
Outline

Why? — Genomic Data + Semanic Web?

Goals — Reshape how genomic data is stored

How? — Ph.D. plan introduction

Impacts — Deliverables and goals



PENGQUIN

PErsoNal **Genome** QUery IN healthcare and clinical practice

Personal Genome Sequence Data

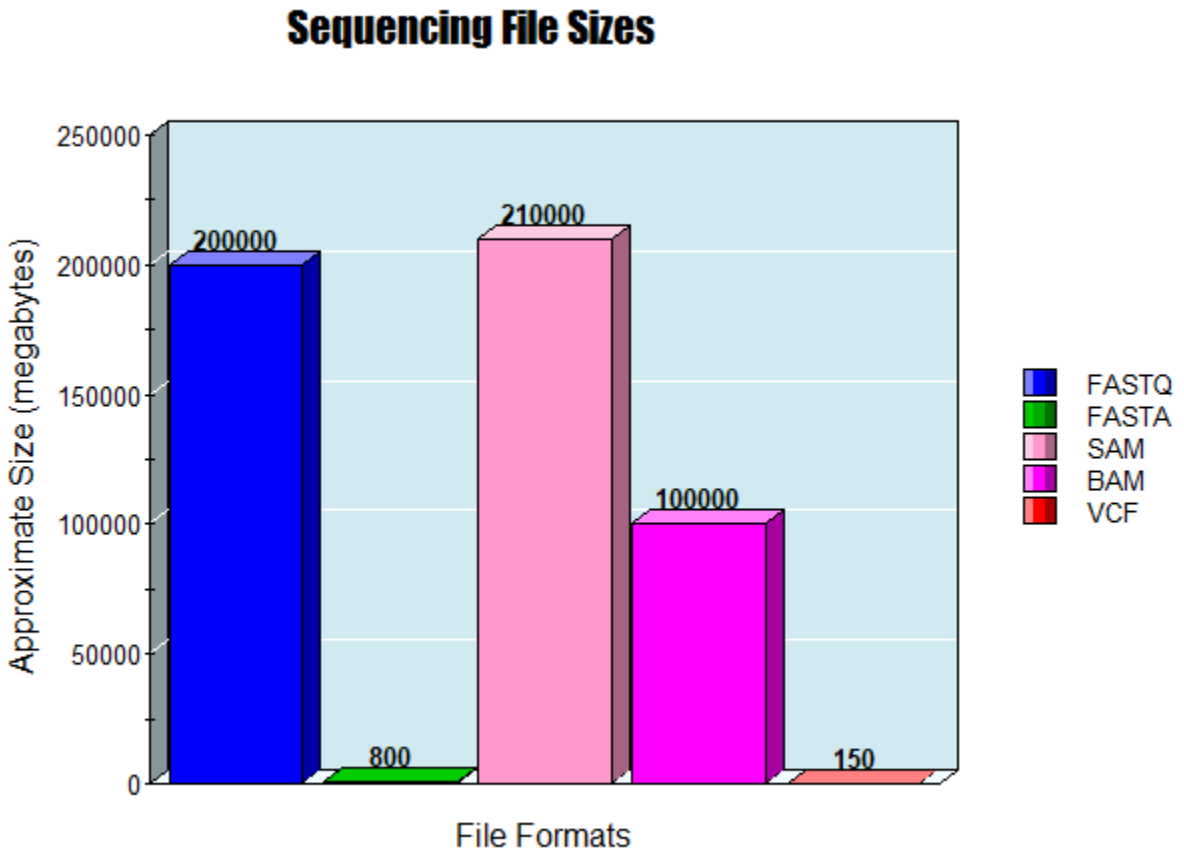
Human genome data is big…

~3.2 BILLION base pairs ----> ~1 GB

Sequencing output file ----> ~200 GB

Variant Call Format (***VCF***) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

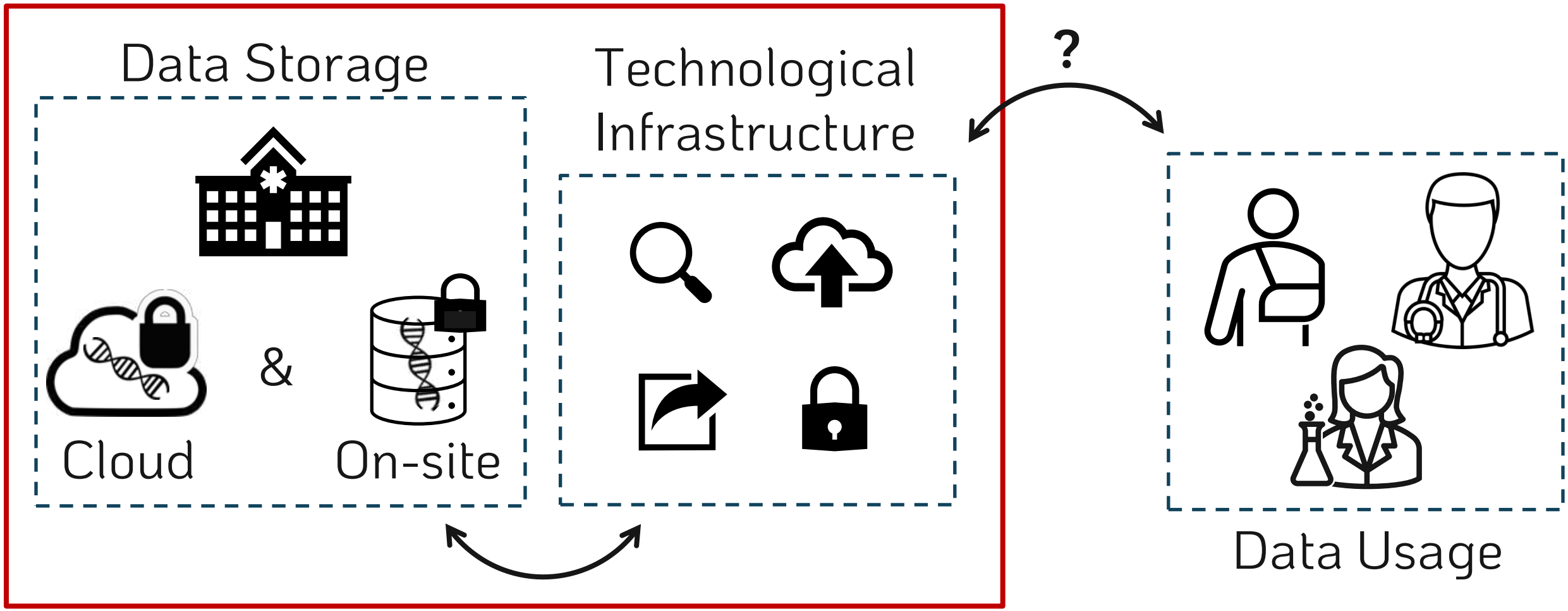


Other considerations

- Storage
- Privacy
- Governance
- Formatting

Bergström et al. SCIENCE 20 Mar 2020.

Semantic Genome Data?



Scaling Genome Data Usage

Research genomic data sharing initiatives:

FHIR

Beacon

OHDSI

OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces

→ Not much...

Solid Technology

- Does not solve ALL problems, but offers an environment where improvements are possible
  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

https://solidproject.org/TR/protocol

Research Questions

1. Can Solid be implemented for genomic data storage?

What is the impact on data policy enforcement?

2. Can (VCF) genomic data be represented as RDF?

Can genomic and other medical data be linked semantically?

3. Can RDF genomic data be queried in a useful way?

Can this improve genomic data discoverability for researchers?

Project Goals

Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

Project Goals

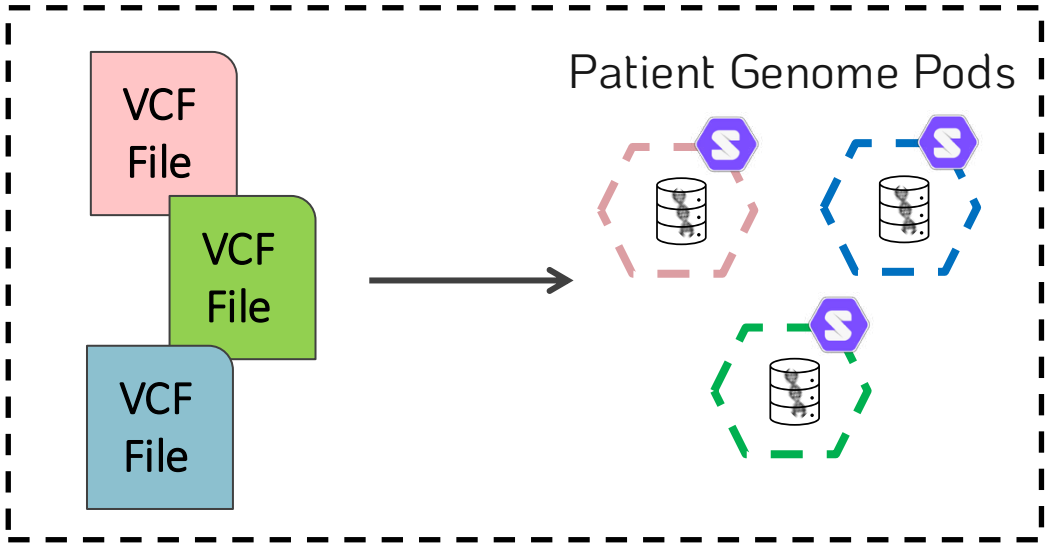
But wait...

This sounds like an **ENGINEERING** project!

Where is the **RESEARCH**?

Workflow Overview

Data Storage

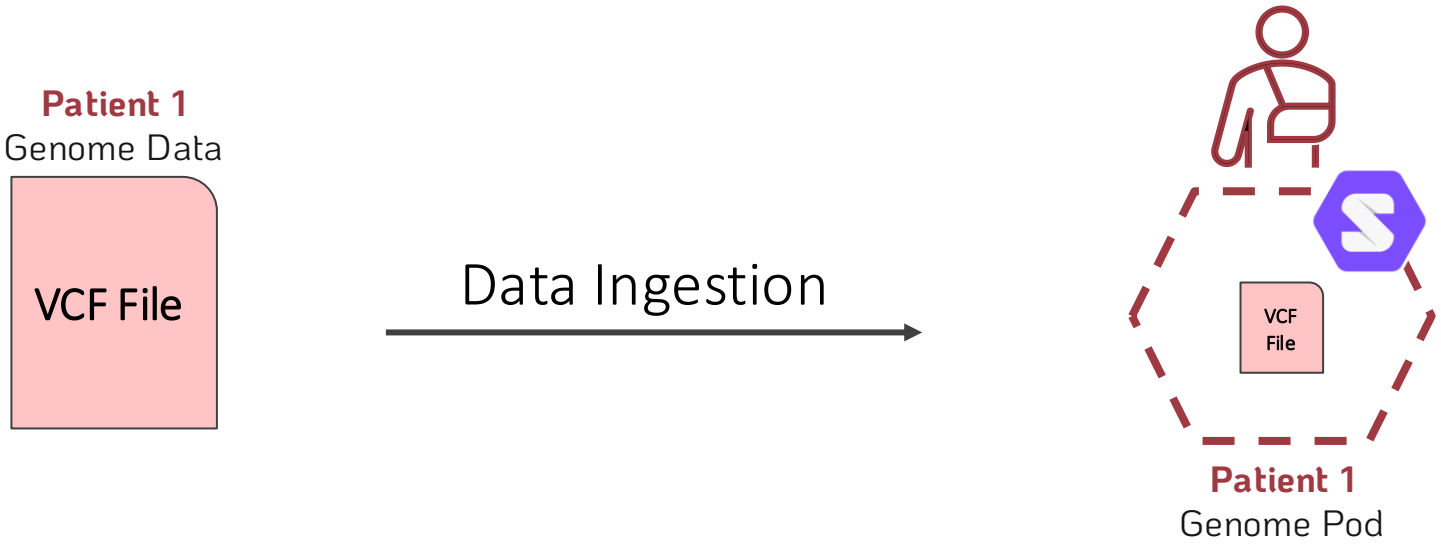


Data Storage

Solid can be implemented for genomic data storage.

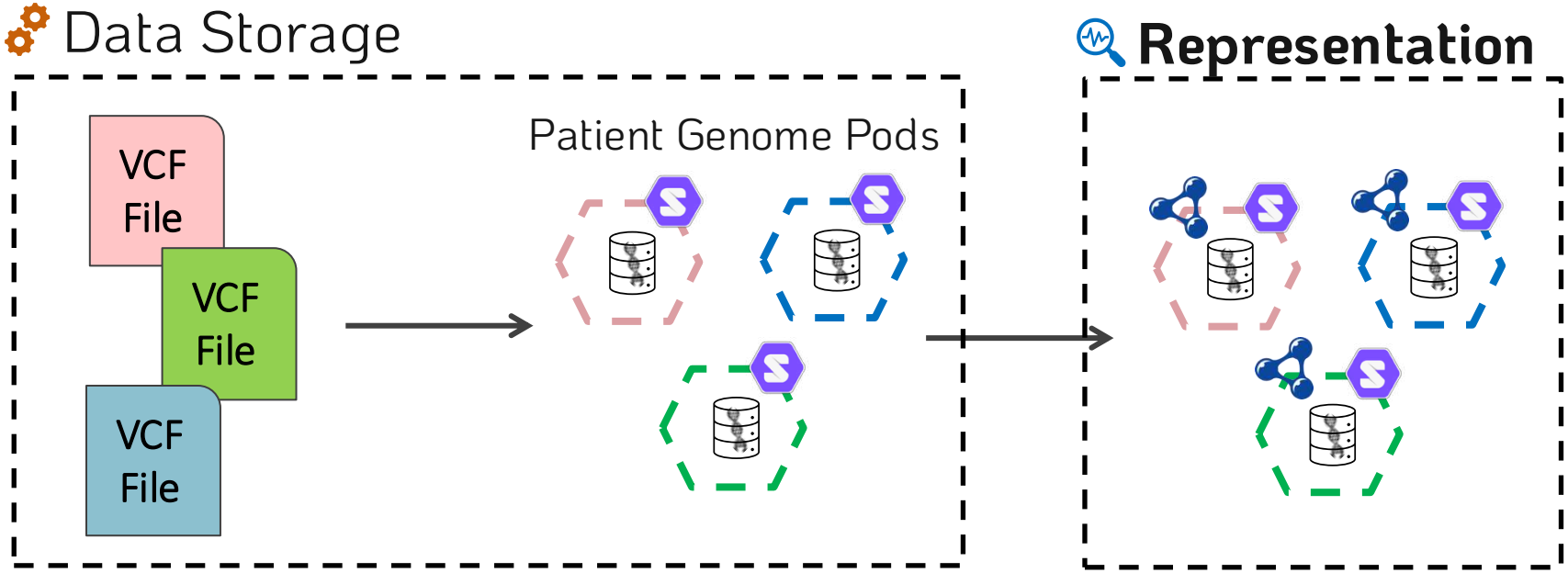
Solid pod instances → Community Solid Server[1]

VCF Genomic Data → Illumina Platinum Genomes[2]



[1] https://github.com/CommunitySolidServer/CommunitySolidServer
[2] https://emea.illumina.com/platinumgenomes.html

Workflow Overview

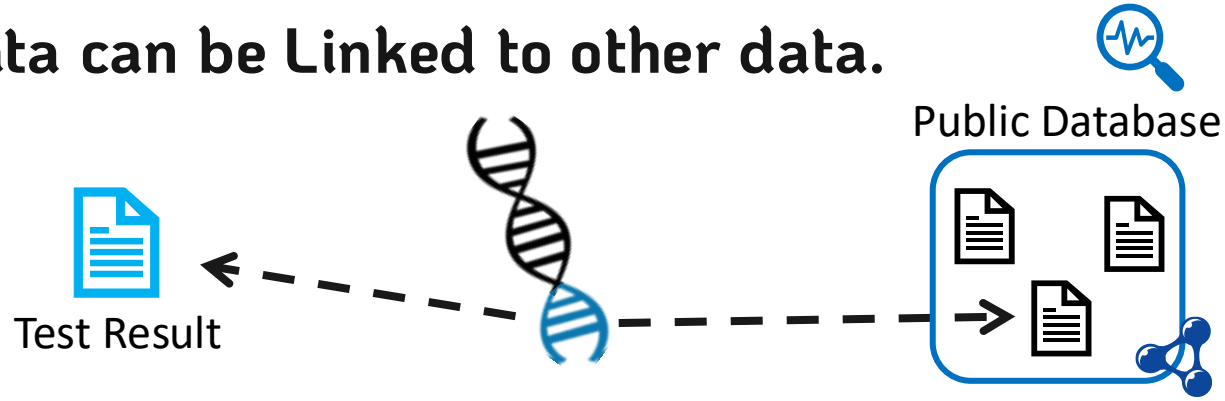


Genome Data in RDF

VCF genomic data can be represented as RDF.

VCF to RDF Vocabularies exist → SPHN Semantic Interoperability Framework[3] + FHIR HL7 Format[4]

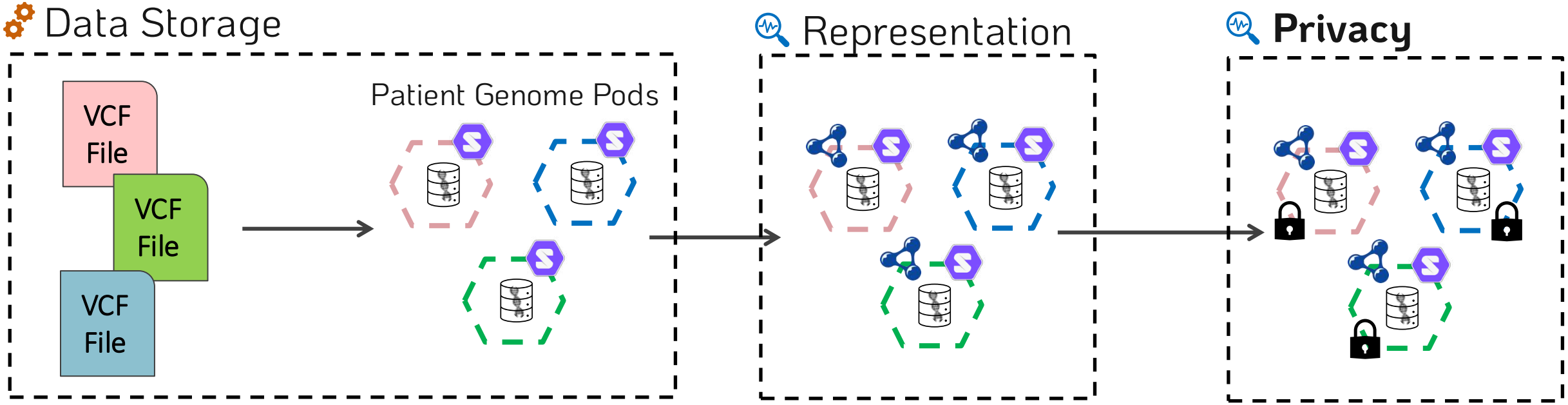
VCF genomic data can be Linked to other data.



How to automate this?
What vocabularies could be used for the linkages?

[3] van der Horst E, et al. Bridging Clinical and Genomic Knowledge: An Extension of the SPHN RDF Schema for Seamless Integration and FAIRification of Omics Data. 2023.
[4] Dolin RH, et al. vcf2fhir: a utility to convert VCF files into HL7 FHIR format for genomics-EHR integration. 2021.

Workflow Overview



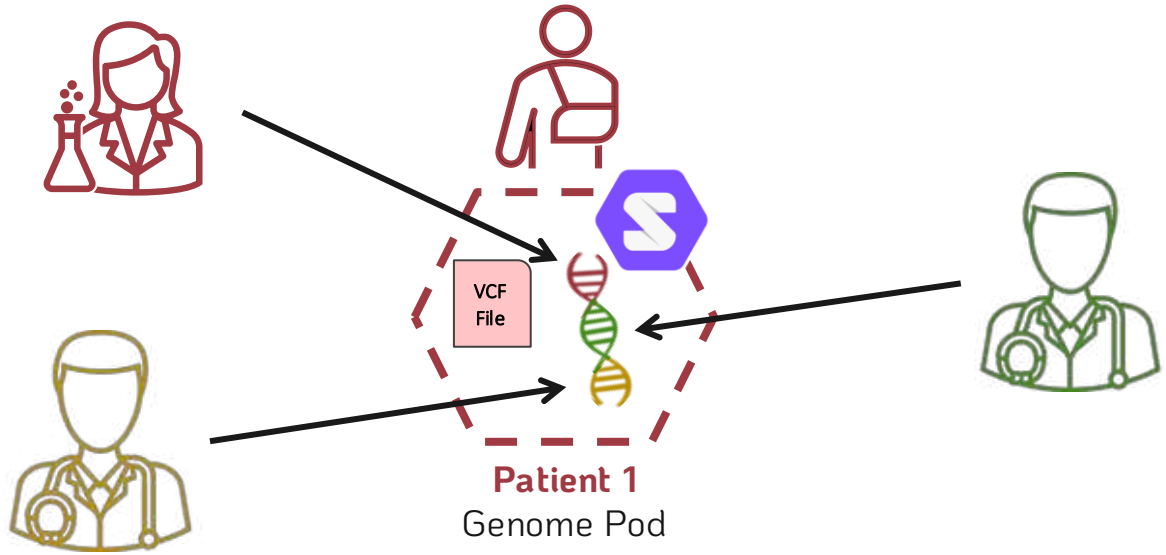
Data Privacy Controls

Granular privacy controls can be implemented in Solid for genomic data.

Web Access Contol → Solid Protocol ACP[5]

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.

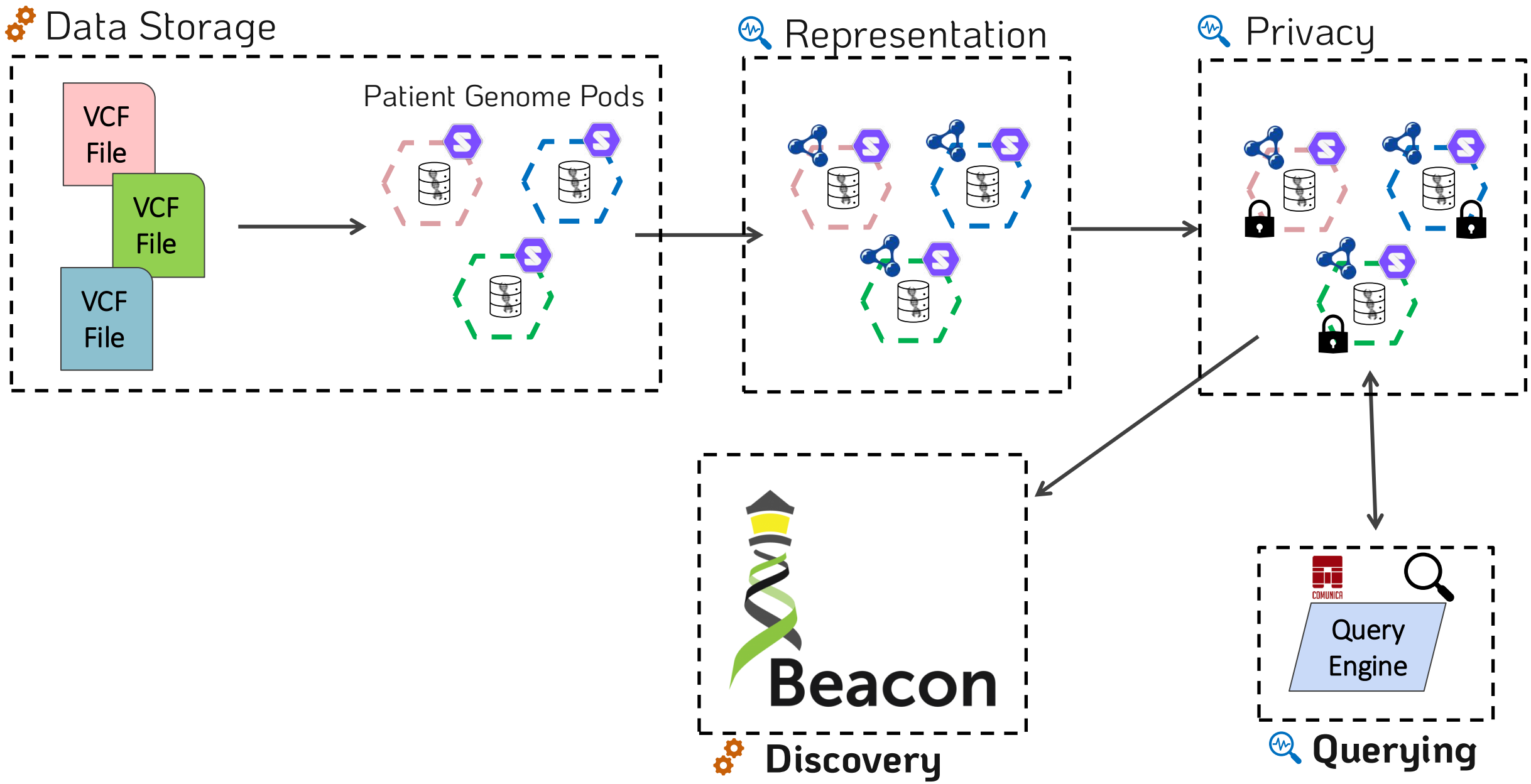


Chromosome is a semantic Class

Mutation is also a semantic Class, above a triple but below a Chromosome

[5] https://solidproject.org/TR/protocol

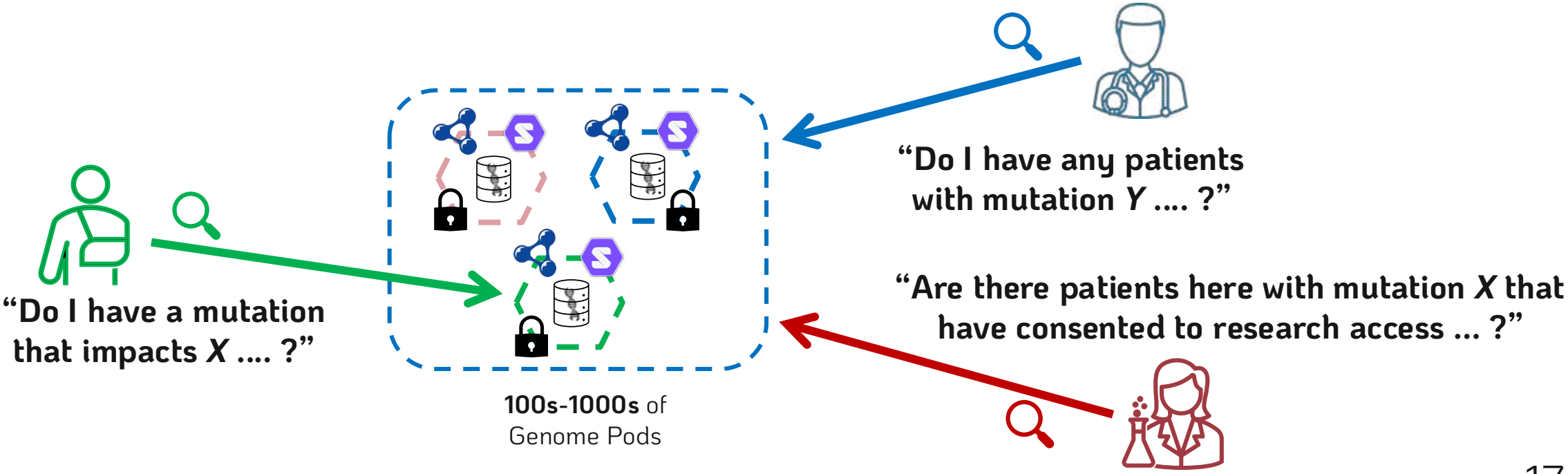
Workflow Overview



Querying Genome Pods

RDF genomic data pods can be queried.

Query Engine → Custom Comunica implementation/engine[6]



[6] https://comunica.dev/

PhD Goals

Finished

Soon

Future

Basic Functionalities:

- Genomic data storage in Solid pods
- Data privacy management
- Data Sharing via the web

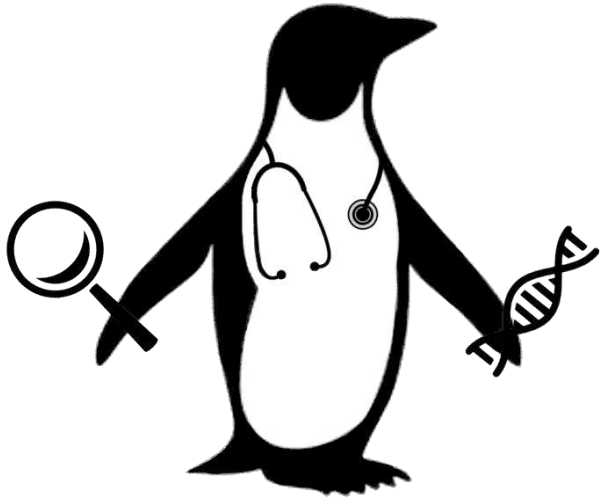
Experimental research:

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?

GHENT UNIVERSITY

IDLab INTERNET & DATA LAB

vito

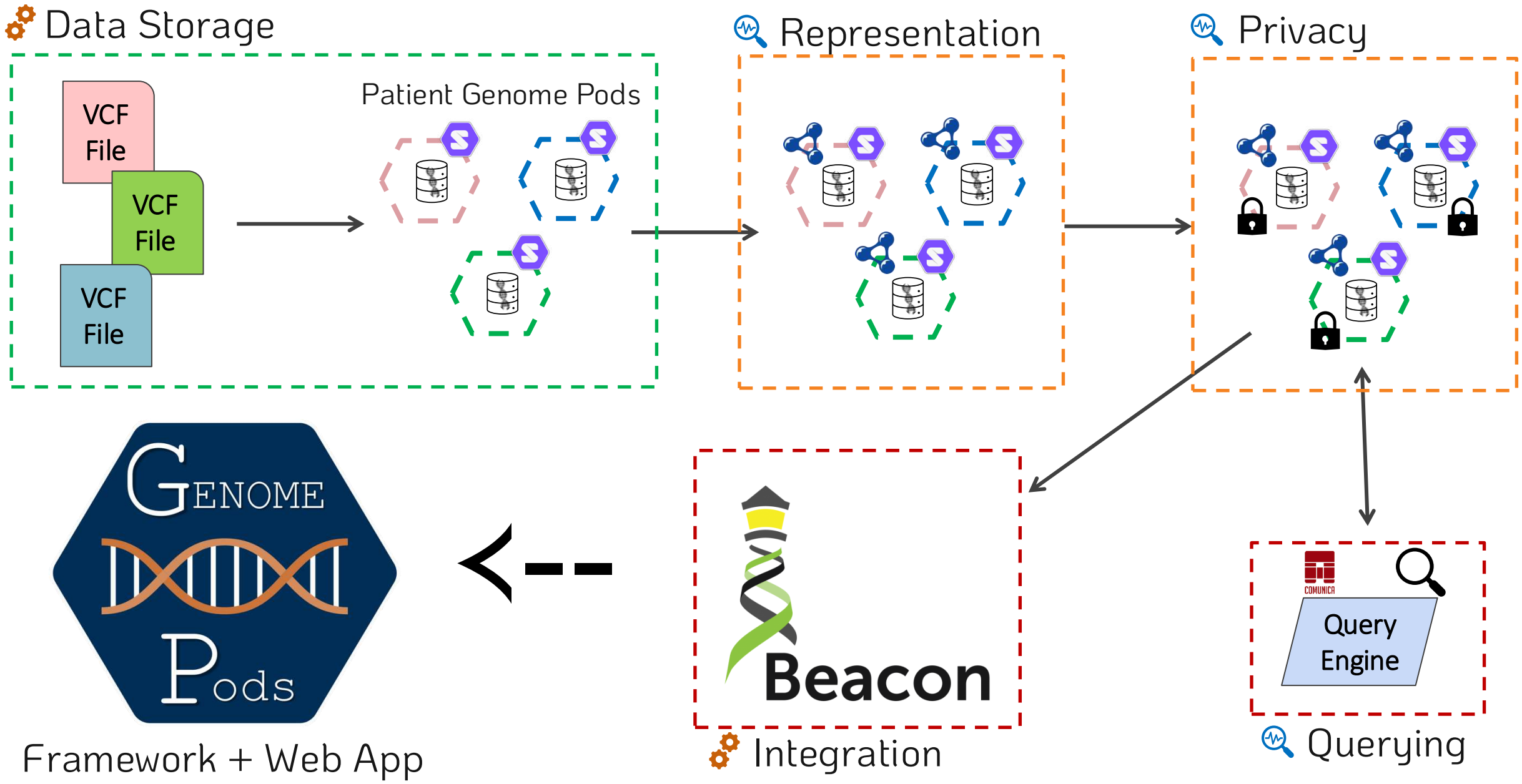


Questions?

Paper

Contact

Workflow Overview



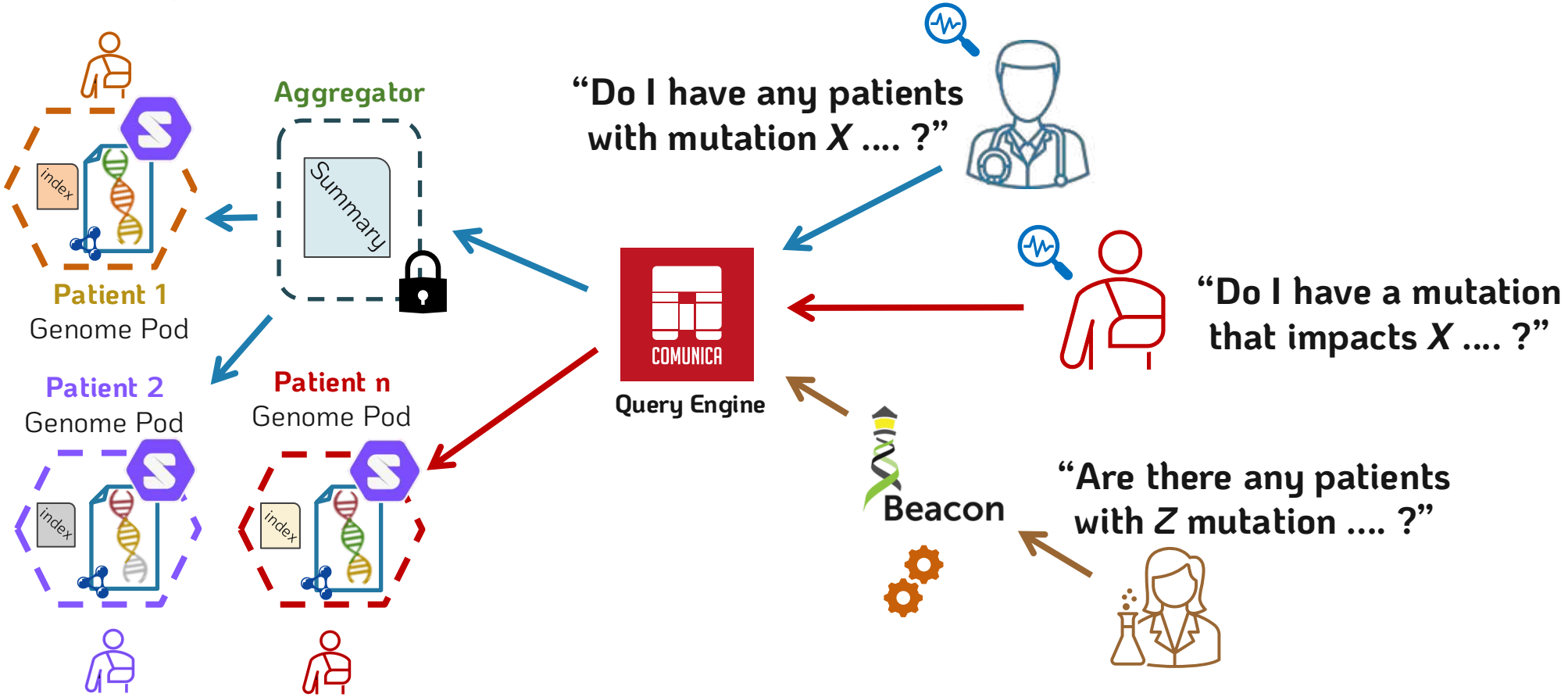
Querying Genome Pods (possibilities)

Single Pod Query

Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed